1 (20 points) TRUE/FALSE

State if the following statements are true or false. Please write TRUE or FALSE in the space provided. Each problem is worth 1 point, but you will be penalized -1 point for an incorrect answer, so leave the answer blank if you don't know.

(a) In a distributed computing cluster, each node typically shares the same CPU and memory with other nodes on the rack.

Answer:_____

(b) If a Map Worker fails, all tasks it has completed must be redone, because its output is stored locally and not in the distributed file system.

Answer:_____

(c) In Spark, the GroupByKey transformation creates a wide dependency because each output partition may depend on multiple input partitions.

Answer:_____

(d) RDDs in Spark can be updated in place to modify existing records.

Answer:_____

(e) The *interest* of an association rule is always positive.

Answer:_____

(f) Market-basket analysis can be applied only to retail data and not to other domains such as student-club relationships.

Answer:_____

(g) The PCY algorithm can efficiently use the triangular-matrix method to count frequent pairs.

Answer:_____

(h) In the multistage algorithm, truly frequent pairs will always hash to frequent buckets across all stages.

Answer:_____

(i) The expected similarity between two minhash signatures equals the Euclidean distance between the original sets.

Answer:_____

(j) Longer minhash signatures lead to smaller expected errors when estimating similarity.

Answer:_____

(k) After minhashing, LSH guarantees that all pairs of similar documents will be identified.

Answer:_____

(l) In the banding technique, signatures are divided into several bands, and each band is hashed independently to identify candidate pairs.

Answer:_____

(m) A clustroid represents an existing data point that is closest to all other points within the same cluster.

Answer:_____

(n) Hierarchical clustering may stop merging clusters when the merged cluster's diameter exceeds a predefined threshold.

Answer:_____

(o) In the BFR algorithm, the discard set contains points that have not yet been assigned to any cluster.

Answer:_____

(p) In the GRGPF algorithm, a new point cannot be the clustroid immediately when it is added.

Answer:_____

(q) The eigenvectors of a symmetric matrix are always orthogonal to each other.

Answer:_____

(r) The first principal component corresponds to the eigenvector associated with the smallest eigenvalue of the covariance matrix.

Answer:_____

(s) The number of non-zero singular values in SVD equals the number of rows in the data matrix.

Answer:_____

(t) The low-rank approximation of a matrix obtained through SVD minimizes the reconstruction error among all matrices of the same rank.

Answer:_____

2 [15 Points] Finding Similar Items

Note: To receive full credit for each sub-question, **all entries must be correct** (no partial points will be given).

(a) [4 points] We want to create the characteristic matrix for given three documents $S1$, $S2$, and $S3$ using k-shingles with $k = 6$. We should normalize each document by converting to lowercase, and removing punctuation before forming 6-shingles.

- $S1$: Hanwha Eagles is the most popular sports club in Korea. To celebrate the playoff events, every seat is decorated with a single flower.
- $S2$: Hanwha Life Esports team always excites the crowds. Their events include tournaments against famous teams.
- $S3$: In October, Daejeon flower festival is held. The main events are flower exhibitions and marathon.

Construct the characteristic matrix by applying indexing function f, from which the following mapping is obtained:

$$\text{hanwha} \to 0, \quad \text{sports} \to 1, \quad \text{events} \to 2, \quad \text{flower} \to 3, \quad \text{crowds} \to 4.$$

(b) [4 points] Suppose we compute minhash signatures for documents as covered in class. Fill in the blanks of the signature matrix shown below using the following hash functions:

$$g_1(x) = 2x - 3 \pmod 5, \qquad g_2(x) = 2x + 3 \pmod 3,$$
$$g_3(x) = 4x + 2 \pmod 3, \qquad g_4(x) = 3x + 4 \pmod 5.$$

Hash fn	Sig(S_1)	Sig(S_2)	Sig(S_3)
g_1			
g_2			
g_3			
g_4			

(c) [3 points] Based on the results in (a) and (b),

1. Compute the true Jaccard similarity for all pairs of sets.
2. Estimate the approximate Jaccard similarities using the minhash signatures.
3. Write any pair of sets whose true and estimated similarities are equal. If no such pair exists, write "none."

(d) [4 points] Let s_{12}, s_{13}, s_{23} be the true Jaccard similarities obtained from parts (c), where s_{ij} denotes the Jaccard similarity between i and j. Consider LSH banding configurations with $m = 4$ rows:

$$(r, b) \in \{(1, 4),\ (2, 2),\ (4, 1)\}$$

1. For each (r, b), write down the expressions for $P_{\text{cand}}(s_{12})$, $P_{\text{cand}}(s_{13})$, and $P_{\text{cand}}(s_{23})$ (numerical computation is not required).

(r, b)	$P_{\text{cand}}(s_{12})$	$P_{\text{cand}}(s_{13})$	$P_{\text{cand}}(s_{23})$
(1,4)			
(2,2)			
(4,1)			

2. Based on your results, simply explain how increasing r affects recall and false positive rate, and explain how increasing b affects recall and false positive rate.

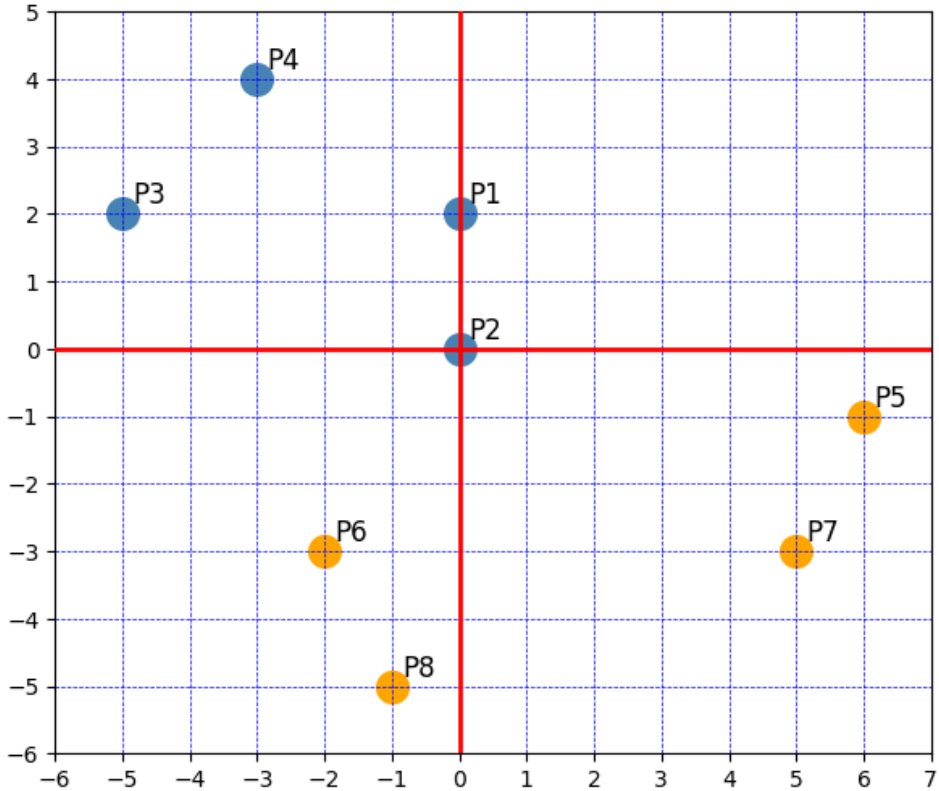
3 [20 points] Clustering

(a) [10 points] For this question, you will simulate the **CURE** algorithm. The data points are already clustered into two clusters $\mathbf{C_1}$ and $\mathbf{C_2}$. For each cluster, there are four data points as follows:

Cluster $\mathbf{C_1}$: $\mathbf{P_1} = (0, 2)$, $\mathbf{P_2} = (0, 0)$, $\mathbf{P_3} = (-5, 2)$, $\mathbf{P_4} = (-3, 4)$

Cluster $\mathbf{C_2}$: $\mathbf{P_5} = (6, -1)$, $\mathbf{P_6} = (-2, -3)$, $\mathbf{P_7} = (5, -3)$, $\mathbf{P_8} = (-1, -5)$

All distances use the **Euclidean (L2)** metric.



(i) [2 points] Find the coordinates for the centroid of each cluster using the given data points.

$\mathbf{C_1}$: _____

$\mathbf{C_2}$: _____

(ii) [2 points] For each cluster, you will select 2 representative points. The first representative is the point farthest from the cluster centroid. The second representative is the point farthest from the first representative. Using the centroids calculated from the previous question, select 2 representative points for each cluster.

$\mathbf{C_1}$: The first rep. is _____ and the second rep. is _____

$\mathbf{C_2}$: The first rep. is _____ and the second rep. is _____

(iii) [3 points] Move each chosen representative 50% toward its cluster's centroid. Then, provide the final moved coordinates.

$\mathbf{C_1}$: _____ and _____

$\mathbf{C_2}$: _____ and _____

(iv) [3 points] A new point $\mathbf{P_9} = (0, -1)$ is given. Using the **CURE** assignment rule, determine which cluster $\mathbf{P_9}$ should be assigned to. Show your intermediate steps to get full credit.

(b) [10 points] **A-CURE (Adjusted CURE)**
A-CURE is a variant of the CURE algorithm. Instead of moving all representative points by a fixed fraction toward the centroid, A-CURE adjusts the shrinkage rate for each representative based on its distance to the cluster centroid.

Let d_i be the distance between a representative point P_i and the centroid c. The adjusted shrinkage ratio is

$$\alpha_i = \frac{1}{d_i},$$

and the moved coordinate is

$$P_i' = (1 - \alpha_i)P_i + \alpha_i\, c.$$

Use the same points $\{P_1, \ldots, P_8\}$ and the Euclidean (L2) distance as in part (a).

(i) [4 points] Move each representative from part (a-ii) toward its cluster centroid using the adjusted shrinkage ratio α_i. Report the final coordinates for the two representatives of each cluster.

$\mathbf{C_1}$: _____ and _____

$\mathbf{C_2}$: _____ and _____

(ii) [6 points] The same new point as in part (a-iv), $P_9 = (0, -1)$, is given. Using the shrunk representatives for each cluster by A-CURE as in part (b-i), determine the cluster assignment for P_9. Show your intermediate steps to get full credit.

4 [20 points] Dimensionality Reduction

Part A. [10 points] There are n students and m clubs. Each student belongs to exactly one club. Let $X \in \mathbb{R}^{n \times m}$ be the *membership matrix* defined by

$$X_{ij} = \begin{cases} 1, & \text{if student } i \text{ is a member of club } j, \\ 0, & \text{otherwise.} \end{cases}$$

Hence, each row of X is a *one-hot* vector indicating the club to which the corresponding student belongs. Let n_j denote the number of students in club j, so that $\sum_{j=1}^{m} n_j = n$. Assume that each club has at least one member, i.e. $n_j > 0$ for all $j = 1, \ldots, m$. Consider the singular value decomposition (SVD) $X = U\Sigma V^\top$ where

$$U \in \mathbb{R}^{n \times m}, \quad V \in \mathbb{R}^{m \times m} \text{ is orthonormal}, \quad \Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_m),$$

and the singular values satisfy $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_m \geq 0$. Here, $\operatorname{diag}(a_1, \ldots, a_m)$ denotes a diagonal matrix with diagonal entries $a_1, \ldots, a_m$ and zeros elsewhere.

(i) [2 points] Express $X^\top X$ in terms of $n_1, \ldots, n_m$.

(ii) [3 points] Express U, Σ, and V in terms of $n_1, \ldots, n_m$.

(iii) [1 point] Let $n_{(1)} \geq n_{(2)} \geq \cdots \geq n_{(m)}$ denote the club sizes sorted in non-increasing order. When only the top $r < m$ singular values of X are retained (rank-r truncated SVD), express the *retained energy ratio* in terms of $n_{(1)}, \ldots, n_{(m)}$.

(iv) [4 points] Let $\mathcal{C}_r$ denote the set of all matrices obtained by selecting **any** r **columns** of X and setting all other columns to zero (i.e., considering only r clubs). Let X_r denote the rank-r truncated SVD of X, i.e., the best rank-r approximation of X in the Frobenius norm. Determine whether there exists $Y^\star \in \mathcal{C}_r$ such that

$$\|X - Y^\star\|_F = \|X - X_r\|_F,$$

and explicitly characterize $Y^\star$ if it exists.

Part B. [10 points] Now assume each student may join multiple clubs, so a row of X can contain more than one "1". Consequently, each row of X is a *multi-hot* vector representing all clubs to which the student belongs. Please answer the following questions.

(i) [2 points] Describe the meaning of each element $(X^\top X)_{jk}$ when students are allowed to join multiple clubs. Explain, in words, what this value tells us about the relationship between clubs j and k.

(ii) [3 points] Assume each club has exactly s members and that any two distinct clubs $j \neq k$ share exactly t members, where $0 < t \leq s$. Express the singular values of X in terms of s, t, and m.

(iii) [5 points] Let $\mathcal{C}_r$ be the set of all matrices obtained by selecting any r columns of X and setting the others to zero, and let X_r denote the rank-r truncated SVD of X as in part A(iv). Assume that each student can belong to multiple clubs, and that the clubs satisfy the *equal-size / constant-overlap* described in part B(ii). Show that for any $r < m$,

$$\|X - X_r\|_F \;<\; \min_{Y \in \mathcal{C}_r} \|X - Y\|_F,$$

i.e., *the rank-r SVD strictly outperforms any column-selection (club-removal) strategy when overlaps exist.*

5 [20 points] Frequent Itemsets

(a) [8 points] Prove or disprove by giving an explicit counterexample.

(i) [2 points] If $X \subseteq X'$, then $\text{conf}(X' \to Y) \leq \text{conf}(X \to Y)$.

(ii) [2 points] If $X \subseteq X'$, then $\text{interest}(X' \to Y) \leq \text{interest}(X \to Y)$.

(iii) [4 points] There exists a market-basket dataset $\mathcal{D}$ consisting of *exactly* 16 *distinct transactions with no duplicates* such that

$$\text{interest}(A \to B) = 0 \quad \text{for } \textit{every } A, B \in \{\, a \mid \text{support}_{\mathcal{D}}(\{a\}) > 0\}, \; A \neq B \; .$$

(Note that A and B are singleton sets.)

(b) [12 points] You have a market-basket dataset $\mathcal{D}$. Let

$$F \;:=\; \{\, X \mid \mathrm{support}_{\mathcal{D}}(X) \geq s \,\}$$

be the frequent itemsets in $\mathcal{D}$, where s is the support threshold. Let $S \subset \mathcal{D}$ be a sample created by keeping each basket independently with probability p.

In this problem, we consider the Toivonen's algorithm not with a single support threshold but using 10 different support thresholds as follows:

$$\tau_k \;:=\; \beta_k p s \quad \text{where } \beta_k = \frac{1}{10}k \quad (k = 1, 2, \ldots, 10)$$

The sample frequent itemsets with threshold τ_k is defined as:

$$F_S^{(k)} \;:=\; \{\, X \mid \mathrm{support}_S(X) \geq \tau_k \,\} \quad \text{for each } k \in \{1, \ldots, 10\}.$$

(i) [1 point] For the same, single sample S, show that the sets of frequent itemsets $F_S^{(k)}$ are nested as follows.

$$F_S^{(10)} \subseteq F_S^{(9)} \subseteq \cdots \subseteq F_S^{(1)}$$

(ii) [3 points] For the same, single sample S, decide whether

$$\mathrm{NB}\big(F_S^{(10)}\big) \;\subseteq\; \mathrm{NB}\big(F_S^{(9)}\big) \;\subseteq\; \cdots \;\subseteq\; \mathrm{NB}\big(F_S^{(1)}\big)$$

must hold where $\mathrm{NB}(F)$ denotes the negative border of F. Prove your claim or disprove it by giving an explicit counterexample.

(iii) [3 points] For the same, single sample S, assume *Strategy 1* uses only $k = 1$ ($\beta_1 = 0.1$), while *Strategy 2* uses only $k = 10$ ($\beta_{10} = 1.0$). Compare and contrast these two strategies with respect to two aspects: (1) the computational cost of Pass 2, and (2) the probability of success.

(iv) [5 points] We define the *success index set* $K_{succ}(S)$ of a sample S as:

$$K_{succ}(S) := \{k \in \{1, \ldots, 10\} \mid \forall X \in \mathrm{NB}(F_S^{(k)}), \mathrm{support}_{\mathcal{D}}(X) < s\}.$$

This set indicates which of the 10 thresholds lead to a successful run of the Toivonen's algorithm. In other words, $k \in K_{succ}(S)$ means that when using the threshold τ_k, its corresponding negative border is "safe," allowing the algorithm to terminate successfully.

Assume that we have drawn two independent samples, S_A and S_B, from $\mathcal{D}$. Decide whether the equality $K_{succ}(S_A) = K_{succ}(S_B)$ must always hold. Prove your claim, or disprove it by providing an explicit counterexample.

6 [25 points] MapReduce and Spark

A quantile is the element at a specific position in an ordered dataset, such as the median. For a dataset $X = \{x_1, \ldots, x_N\}$ of distinct values, define the rank of an element x as the number of elements which is less than or equal to x: $\texttt{rank}_X(x) = |\{x_i \in X : x_i \leq x\}|$. We say x is an ϕ-quantile of X if $x \in X$ and $\texttt{rank}_X(x) = \lceil \phi N \rceil$ where $0 < \phi \leq 1$. Here, $\lceil a \rceil$ returns the smallest integer greater than or equal to a.

(a) [1 point] Compute the 0.5-quantile of $X = \{11, 21, 24, 61, 81, 39, 89, 56, 12, 51\}$.

(b) [5 points] We want to compute a quantile over a large collection of numbers in a distributed way. This can be achieved with a *decomposable* function; a function f over a set X is said to be *decomposable* if there exists a function g such that $f(X) = g(f(X_1), f(X_2))$ for all disjoint subsets X_1, X_2 with $X = X_1 \cup X_2$ and $X_1 \cap X_2 = \emptyset$. Show that the 0.5-quantile of a set X is **NOT** decomposable; provide a specific counterexample.

Hint: Find X, Y with partitions $X = X_1 \cup X_2$, $Y = Y_1 \cup Y_2$ such that $f(X) \neq f(Y)$ but $f(X_1) = f(Y_1)$ and $f(X_2) = f(Y_2)$, thereby showing that no such g can exist.

(c) [4 points] To make the quantile computation decomposable, we define *ϵ-approximate ϕ-quantile* of X as a value $x \in X$, where $\epsilon \ll 1$ is small and $|X| = N$, satisfying

$$|\,\texttt{rank}_X(x) - \phi N| \leq \epsilon N.$$

This allows an error of at most ϵN compared to the ϕ-quantile of X. Let X be as defined in (a). **Choose five elements in** X that cover all 0.1-approximate ϕ-quantiles of X for $\phi \in (0, 1]$. (That is, there exists a query function $q : (0, 1] \to \hat{X}$ returning the 0.1-approximate ϕ-quantile of X where $\hat{X} \subset X$ with $|\hat{X}| = 5$.)

(d) [5 points] As in (c), all ϵ-approximate quantiles can be answered using a subset of elements. We define an *ϵ-approximate quantile summary* of X as

$$S := \{s_1, \cdots, s_M\} \subset X$$

where there exist functions $\texttt{rmin}_X, \texttt{rmax}_X : S \to \mathbb{N}$ such that $s_1 \leq \cdots \leq s_M$, and

- $\texttt{rmin}_X(s_i) \leq \texttt{rank}_X(s_i) \leq \texttt{rmax}_X(s_i)$ for all $1 \leq i \leq M$.
- $\texttt{rmax}_X(s_{i+1}) - \texttt{rmin}_X(s_i) \leq 2\epsilon|X|$ for all $1 \leq i < M$.

By bounding the $\texttt{rank}_X$ between $\texttt{rmin}_X$ and $\texttt{rmax}_X$, each element s_i covers a range of quantiles in X. Let the summary function $S(\cdot)$ map a dataset X to its ϵ-approximate quantile summary S. Then, this summary function is decomposable: there exists a combine function c such that $S(X) = c(S(X_1), S(X_2))$ for all disjoint subsets X_1, X_2 with $X = X_1 \cup X_2$ and $X_1 \cap X_2 = \emptyset$. Design such a function c; no proof is required.

Hint: Let $S(X_1) = \{p_1, \cdots, p_{M_1}\} \subset X_1$ and $S(X_2) = \{q_1, \cdots, q_{M_2}\} \subset X_2$. Then design $S(X) = \{s_1, \cdots, s_M\} \subset X$ with $M = M_1 + M_2$, by defining $\texttt{rmin}_X$ and $\texttt{rmax}_X$ in terms of $\texttt{rmin}_{X_1}, \texttt{rmax}_{X_1}, \texttt{rmin}_{X_2}$, and $\texttt{rmax}_{X_2}$.

In Spark, the ϵ-approximate ϕ-quantile can be computed in a distributed manner:

```
S = sc.parallelize(X) \
    .mapPartitions(lambda Xi: summary(Xi, eps)) \
    .treeReduce(lambda Sj, Sk: merge(Sj, Sk, eps))

result = query(S, phi, eps)         # problem (g)
```

Here, `mapPartitions` creates a summary S_i for each partition $X_i \subset X$, and `treeReduce` recursively merges S_j and S_k into a new summary S_l. The functions are defined as follows. Note that `floor(a)` returns the greatest integer less than or equal to `a`.

```
def summary(Xi, eps):
    Ni = len(Xi); Si = []
    for r, x in enumerate(sorted(Xi)):
        t = {'value': x, 'rmin': r+1, 'rmax': r+1}
        if (r % floor(2*eps*Ni) == 0) or (r == Ni - 1):
            Si.append(t)
    return Si

def merge(Sj, Sk, eps):
    Nl = Sj[-1]['rmax'] + Sk[-1]['rmax']
    Sl = combine(Sj, Sk)            # problem (e)
    Sl = compress(Sl, Nl, eps)      # problem (f)
    return Sl
```

(e) [6 points] In the Map stage, `summary` generates an ϵ-approximate quantile summary S_i for each partition X_i. In the Reduce stage, given summaries S_j and S_k for X_j and X_k, respectively, `combine` should produce an ϵ-approximate quantile summary S_l for $X_j \cup X_k$. Fill (1) and (2) in the code below to satisfy this property.

```
def combine(Sj, Sk):
    Z = sorted(Sj + Sk, key=lambda s:  s['value']); Sl = []
    for z in Z:
        Sz, Sn = (Sj, Sk) if z in Sj else (Sk, Sj)
        prev = max([t for t in Sn if t['value'] < z['value']],
                key=lambda t:  t['value'], default=None)
        next = min([t for t in Sn if t['value'] > z['value']],
                key=lambda t:  t['value'], default=None)
        rmin = [                (1)                ]
        rmax = [                (2)                ]
        t = {'value':z['value'], 'rmin':rmin, 'rmax':rmax}
        Sl.append(t)
    return Sl
```

(f) [2 points] In `compress`, we make the summary compact by ensuring that

$$\texttt{rmax}_X(v_{i+2}) - \texttt{rmin}_X(v_i) > 2\epsilon|X|$$

for all $1 \leq i < s - 1$. Fill (3) in the code below so that this condition is satisfied.

```
def compress(S, N, eps):
    R = [S[0]]; prev = S[0]; T = 2*eps*N
    for t in S[1:]:
        if [                (3)                ]:
            R.append(prev)
        prev = t
    R.append(S[-1])
    return R
```

(g) [2 points] We can design `query(S, phi, eps)` that returns an ϵ-approximate ϕ-quantile of X, given an ϵ-approximate quantile summary of X. Fill (4) in the code below.

```
def query(S, phi, eps):
    N = S[-1]['rmax']; target = phi*N; tol = eps*N
    for i in range(len(S)):
        if [                (4)                ]:
            return S[i]['value']
    return S[-1]['value']
```